

NATIONAL UNIVERSITY OF MODERN LANGUAGES

Faculty of Engineering and Computing
Department of Software Engineering



Lab Task

DBMS Lab

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Single Row Functions

1. **UPPER(first_name):** Converts first_name to uppercase.
2. **LOWER(last_name):** Converts last_name to lowercase.
3. **SUBSTR(email, 1, 10):** Extracts a substring from email, starting at position 1 for 10 characters.
4. **LENGTH(first_name):** Returns the number of characters in first_name.
5. **TRIM(email):** Removes leading and trailing spaces from email.
6. **ROUND(salary):** Rounds the salary to the nearest integer.
7. **TRUNC(salary):** Truncates salary to an integer (removes decimal places).
8. **MOD(salary, 1000):** Returns the remainder of salary when divided by 1000.
9. **SYSDATE:** Retrieves the current date and time.
10. **ADD_MONTHS(hire_date, 12):** Adds 12 months to hire_date.
11. **MONTHS_BETWEEN(SYSDATE, hire_date):** Calculates the number of months between the current date and hire_date.
12. **TRUNC(hire_date):** Truncates hire_date to remove the time component.
13. **TO_CHAR(hire_date, 'YYYY-MM-DD'):** Converts hire_date to a string in 'YYYY-MM-DD' format.
14. **TO_NUMBER('150000.50'):** Converts the string '150000.50' to a numeric value.

```
SELECT employee_id,
       UPPER(first_name) AS upper_first_name,
       LOWER(last_name) AS lower_last_name,
       SUBSTR(email, 1, 10) AS email_substr,
       LENGTH(first_name) AS first_name_length,
       TRIM(email) AS trimmed_email,
       ROUND(salary) AS rounded_salary,
       TRUNC(salary) AS truncated_salary,
       MOD(salary, 1000) AS salary_modulus,
       SYSDATE AS current_date,
       ADD_MONTHS(hire_date, 12) AS hire_date_plus_one_year,
       MONTHS_BETWEEN(SYSDATE, hire_date) AS months_since_hired,
       TRUNC(hire_date) AS truncated_hire_date,
       TO_CHAR(hire_date, 'YYYY-MM-DD') AS hire_date_str,
       TO_NUMBER('150000.50') AS numeric_salary
FROM ahmad;
```

EMPLOYEE_ID	UPPER_FIRST_NAME	LOWER_LAST_NAME	EMAIL_SUBSTR	FIRST_NAME_LENGTH	TRIMMED_EMAIL	ROUNDED_SALARY	TRUNCATED_SALARY
1	IJLAL	ahmad	ijlalahmad	5	ijlalahmad552@gmail.com	155000	155000
2	6 FAIZAN	khan	faizan1212	6	6 faizan1212@gmail.com	140000	140000

TRUNCATED_SALARY	SALARY_MODULUS	CURRENT_DATE	HIRE_DATE_PLUS_ONE_YEAR	MONTHS_SINCE_HIRED	TRUNCATED_HIRE_DATE
1	155000	0 26-OCT-24	10-JAN-24	21.52031025985663082437275985663082437276	10-JAN-23
2	140000	0 26-OCT-24	20-JUN-23	28.19772961469534050179211469534050179211	20-JUN-22

TE	HIRE_DATE_STR	NUMERIC_SALARY
	2023-01-10	150000.5
	2022-06-20	150000.5